Zeros of Polynomial Functions (2.3) Name ____________________

Circuit

Begin with the problem in the top left. Answer the question. To advance in the circuit, find your answer and write "2" in the blank. Solve that problem, find your answer and write "3" in the blank and so on until you have solved every problem and returned to the beginning.

ANSWER: $\{-3,\ 1\}$ # ______ Find the zeros of $f(x) = x^2 - x - 6$.	ANSWER: $(x-1)(x+4)$ # ______ State the multiplicity of $x = 0$ in $f(x) = x^3(x-2)$.
ANSWER: yes # ______ Find the zeros of $f(x) = x(x-5)(x+1)$.	ANSWER: no # ______ Write the factor corresponding to a zero at $x = -2$.

ANSWER: $\{-2,\ 3\}$

________ Is $x = 2$ a zero of $f(x) = x^2 - 4$?

ANSWER: $\{-4,\ 4\}$

________ Is $x = 1$ a zero of $f(x) = x^2 - 4$?

ANSWER: $(1, 2)$

________ Write the factors of a polynomial with zeros 1 and -4.

ANSWER: $(x + 2)$

________ Find the zeros of $f(x) = (x - 1)^2(x + 3)$.

ANSWER: $\{-1,\ 0,\ 5\}$

________ By the IVT, if $f(1) = -2$ and $f(2) = 3$, a zero lies in which interval?

ANSWER: 3

________ Find the zeros of $f(x) = x^2 - 16$.